

PROBLEM STATEMENT #23 | Smart Cities, Transport & Logistics

Hyperlocal Courier Dispatch & Tracking Engine

1. Problem Context & Overview

An on-demand package delivery platform that assigns local parcel pickups to nearest delivery riders, optimizes multi-stop routing, and enforces OTP verification at the destination.

Target Stakeholders / Actors: *Sender Client, Delivery Rider*

2. Sample Functional Requirement (FR) Guideline

Sample Requirement: FR-001 [Priority: High]

Description: The system shall match an outgoing courier request with the nearest active delivery rider within a 3 km radius.

Sample Acceptance Criteria: Pass: Rider receives dispatch notification with pickup location; Fail: Order assigned to offline rider.

3. Sample Non-Functional Requirement (NFR) Guideline

Sample Requirement: NFR-001 [Type: Performance & Security]

Description: Delivery status and live GPS telemetry updates from riders shall transmit to the sender client with under 2-second latency.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]

- **Complete Requirements Table:**

- >> Exactly 5 Functional Requirements (FR-001 to FR-005)
- >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

- **UML Use-Case Diagram:**

- >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

- **Use-Case Flow Specification:**

- >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
- >> One Alternate Flow.